Aadvik Mishra

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Education

University of Massachusetts, Amherst *Expected Graduation: May 2026*
Bachelor of Science in Computer Engineering GPA: 3.6

Relevant Coursework: VLSI Design, Embedded Systems, Digital Systems, Computer Architecture, Signals & Systems

Awards & Grants: Chancellor's Award, Dean's List

Technical Skills

EDA & Physical Design: Cadence (Innovus, Virtuoso, Pegasus), Synopsys Design Compiler, Synopsys HSPICE, Icarus Verilog
VLSI & Verification: Synthesis, place & route, DRC/LVS, parasitic extraction (PEX), static timing analysis, mixed-level simulation
Embedded & Firmware: C, C++, Embedded C, Python, Bash, MATLAB, real-time firmware, sensor integration, motor control
Hardware & PCB: KiCad, schematic capture, component selection, BOM development, board bring-up, signal integrity analysis
Protocols: I2C, SPI, UART, ADC, PWM, GPIO, TCP/IP
Platforms: ARM Cortex-A9, ATtiny85, ESP32, Arduino, DE1-SoC FPGA
Tools: Git, Linux, Docker, JIRA, Agile, ModelSim, Quartus Prime

Work Experience

Embedded Systems & Controls Engineer Jan. 2025 - Present
UMass Mechatronics Team – ASME Student Design Competition
- Developed C/C++ firmware for real-time motor control and sensor fusion
- Diagnosed hardware-software faults during board bring-up and field testing alongside mechanical and electrical teams.
- Integrated sensors across I2C, SPI, and UART interfaces
- Validated timing and signal integrity using oscilloscopes and logic analyzers.

Undergraduate Course Assistant Jan. 2024 – Dec. 2024
UMass ECE Department
- Mentored 80+ students in digital logic, Verilog RTL design, and FPGA simulation
- Delivered technical training & guided structured debugging & hardware verification techniques across a diverse student base.
- Developed clear technical explanations for complex hardware concepts

Software Engineering Intern June 2025 – Aug. 2025
WealthTechAI
- Performed system-level validation on Linux-based platforms
- Investigated failures through structured root-cause analysis tracing issues across process execution and system state.
- Built Python-based automation for regression testing and functional verification
- Documented system behavior and edge cases to support reliable integration.

Research Assistant Dec. 2023 – Aug. 2024
Elaine Marieb Center for Nursing and Engineering Innovation
- Assembled and debugged PCB-based embedded systems for biomedical sensing devices
- Wrote C and Python test utilities to validate system behavior and data integrity.
- Automated validation workflows and established repeatable testing procedures for cross-team traceability.

Projects

EchoSafe – Embedded Consumer Device (Capstone) | KiCad, Embedded C, Python, ML Present
- Owned full hardware lifecycle: schematic capture, PCB layout, component selection, BOM development, vendor coordination, board bring-up, and stress testing.
- Implemented embedded C firmware for real-time sensor processing and microcontroller-based control on a custom PCB.
- Trained and deployed an on-device sound classification ML model for real-time assistive alerting for hearing-impaired users.

Bias Detection in Automated Hiring | Python, OpenAI API, TF-IDF, Statistical Analysis May. 2025
- Built LLM-based automation pipelines using the OpenAI API to process 480+ structured datasets
- Demonstrated practical ability to leverage AI models for data analysis and automated decision-making workflows.

3×3 Mesh Network-on-Chip (NoC) | Cadence Innovus, Synopsys Design Compiler, Icarus Verilog Dec. 2025
- Synthesized and ran place & route in Cadence Innovus
- Validated deadlock-free routing via mixed-level simulation combining post-synthesis netlists with RTL tiles.